CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination (EC/CE/IT)

Nov-Dec. 2013

EE103: BASICS OF ELECTRONICS & ELECTRICAL ENGINEERING

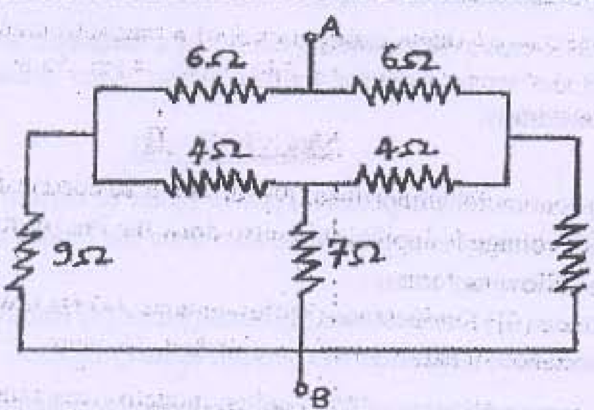
Date: 05/12/2013 **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

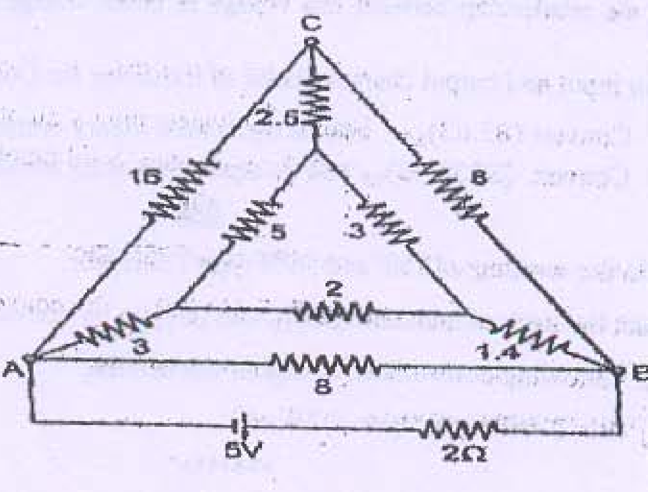
Q - 1 (a) Find the resistance between the terminal A and B in the network shown in Figure. **[04]**



(b) Explain the difference between following terms : **[03]**

(i) Branch and Loop
(ii) Node and Junction
(iii) Network and mesh

Q - 2 (a) With the help of Star-delta Transformation find the current supplied by Battery in the following network. **[07]**



(b) Give the Comparison between Electric Circuit and Magnetic Circuit **[07]**

OR

Q - 2 (a) What is Capacitor? Derive the expression for the equivalent capacitance of capacitors connected in parallel and in series. [07]

(b) With Reference to electrostatic and capacitance:
Define: State Coulomb's Laws, Electric field Intensity, Electric potential, mutual induction, Permittivity, and Capacitance. [07]

Q - 3 (a) Derive emf equation of AC current [07]

(b) Coil A and B with 50 and 500 Turns respectively are wound side by side on a closed iron circuit of section 50 cm^2 and mean length of 1.2m. Estimate (i) Self Inductance of each coil, (ii) Mutual Inductance between coils, (iii) emf induced in coil A if the current in coil B grows steadily from 0 to 5 Amp in 0.01 Sec. Assume relative permeability of iron as 1000. [07]

OR

Q - 3 (a) Explain Hysteresis loss and Eddy current loss in detail with neat diagram. [07]

(b) Draw the B-H curve (Magnetizing curve) of a magnetic material and label the figure. Define all the terms relevant to this curve. Also draw B-H curve for different magnetic materials. [07]

SECTION – II

Q - 4 (a) Derive expressions for impedance, power factor & current for an R-L-C series circuit when an A.C voltage is applied to it. also draw the Phasor diagram. [04]

(b) Explain the following terms :
(1) Admittance (2) Conductance (3) Susceptance (4) Bandwidth [03]

Q - 5 (a) Explain resonance in parallel RLC circuit & its Q-factor. [07]

(b) A coil having a resistance of 27 Ω and an inductive reactance of 32Ω is connected to a capacitive reactance of 44Ω. Find the impedance, cuircuit current and power consumed. Supply voltage =200V [07]

OR

Q - 5 (a) Define the efficiency of full wave rectifier circuit with neat diagram. [07]

(b) Derive the relationship between line voltage & phase voltage in a 3-phase star-connected circuit. [07]

Q - 6 (a) Explain input and output characteristics of transistor for Common-emitter connection. [07]

(b)
1. Convert $(85.63)_{10}$ into its equivalent binary numbers.
2. Convert $(3000.45)_{10}$ into its equivalent octal number. [07]

OR

Q - 6 (a) Explain the working of PNP and NPN type Transistor. [07]

(b) Subtract the decimal numbers $(38)_{10}$ and $(29)_{10}$ by converting them into binary. [03]

(c) Define following terms related to digital electronics. [04]

1) Bit 2) Nibble 3) Byte 4) Word

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